

Quantum Field Theory

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Syllabus (Two Semesters)

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Course Information

- **Primary texts:** Tom Lancaster and Stephen J. Blundell, *Quantum Field Theory for the Gifted Amateur* (L&B, 50 chapters); Abrikosov, Gor'kov, and Dzyaloshinskii, *Quantum Field Theoretical Methods in Statistical Physics* (AGD, 42 sections).
- **Prerequisites:** undergraduate quantum mechanics, classical mechanics, basic special relativity, Fourier transforms, and comfort with Lagrangian methods. L&B Ch.0 reviews the essentials.
- **Duration:** approximately 26 weeks. Semester A covers QFT fundamentals through Feynman diagrams; Semester B covers path integrals, renormalization, spin, QED, many-body applications, and gauge theory.
- **Reading key:** [L&B Ch.X] denotes Lancaster and Blundell; [AGD Section X] denotes Abrikosov, Gor'kov, and Dzyaloshinskii; bracketed “Next” entries denote follow-up texts.

1 Semester A: Fundamentals and Feynman Diagrams

1.1 Phase 1: The Classical Foundation

Lagrangians, classical fields, and special relativity review. L&B Parts I–II; AGD not yet needed. This phase covers Weeks 1–2.

1.1.1 Week 1: Overture, Lagrangians, and the Principle of Least Action

- L&B Ch.0 reviews what QFT is, fields, special relativity, and Fourier transforms. Ch.1 covers Fermat's principle, Newton's laws from Lagrangians, functionals, and least action. Ch.2 develops the simple harmonic oscillator, the continuum limit, and phonons.

Application. Every quantum field is a harmonic oscillator. If you understand the SHO deeply, you understand the vacuum.

Readings. [L&B Ch.0–2]

1.1.2 Week 2: Occupation Numbers, Second Quantization, Continuous Systems

- L&B Ch.3 introduces occupation number representation, creation and annihilation operators, indistinguishability, and the continuum limit. Ch.4 develops field operators, second-quantizing operators, kinetic energy, tight-binding Hamiltonians, and the Hubbard model. Ch.5 introduces Lagrangian and Hamiltonian densities, charged particles in electromagnetic fields, and classical fields.

Application. Second quantization is the language of many-body physics. Bosons and fermions differ by statistics, not by being different kinds of particles.

Readings. [L&B Ch.3–5]

Checkpoint. Note on pacing: L&B chapters are short. Aim for steady reading and work the chapter-end exercises; they are short and diagnostic.

1.2 Phase 2: Writing Down Lagrangians and Canonical Quantization

Klein–Gordon equation, Noether's theorem, canonical quantization, gauge fields. L&B Parts II–III. This phase covers Weeks 3–5.

1.2.1 Week 3: Relativistic QM, Lagrangians for Fields, Symmetry

- L&B Ch.6 introduces the Klein–Gordon equation, probability currents, and negative-energy states. Ch.7 gives Lagrangians for massless and massive scalar fields, ϕ^4 theory, two scalar fields, and complex scalar fields. Ch.8 compares Schrödinger and Heisenberg pictures, time evolution, and the failure of single-particle relativistic QM.

Application. The Klein–Gordon equation is what happens when one applies QM to a relativistic particle too naively; its failure motivates fields.

Readings. [L&B Ch.6–8]

1.2.2 Week 4: Symmetry, Noether’s Theorem, and Canonical Quantization

- L&B Ch.9 covers quantum mechanical transformations, translations, rotations, representations, and Lorentz transformations. Ch.10 covers invariance, conservation, Noether’s theorem, spacetime translation, and the energy–momentum tensor. Ch.11 introduces canonical quantization of fields, normalizing factors, Hamiltonians, normal ordering, and mode expansions.

Application. Every conservation law is a symmetry. Noether’s theorem is one of the most important results in physics.

Readings. [L&B Ch.9–11]

1.2.3 Week 5: Examples of Quantization, Gauge Fields, Discrete Symmetries

- L&B Ch.12 treats canonical quantization of the complex scalar field, Noether currents, and the non-relativistic limit. Ch.13 covers fields with many components, massive electromagnetism, internal symmetries, and polarizations. Ch.14 introduces gauge fields and electromagnetism as the simplest gauge theory. Ch.15 covers charge conjugation, parity, and time reversal.

Application. Gauge invariance is the deepest principle in the Standard Model. Every force is a gauge theory.

Readings. [L&B Ch.12–15]

Checkpoint. Checkpoint: write down scalar/vector-field Lagrangians, apply Noether’s theorem, and canonically quantize the free scalar field.

1.3 Phase 3: Propagators, Perturbation Theory, and Feynman Diagrams

The diagrammatic machinery of QFT. L&B Part IV; this is the payoff of the first five weeks. This phase covers Weeks 6–9.

1.3.1 Week 6: Propagators and Green’s Functions

- L&B Ch.16 introduces Green’s functions, propagators in quantum mechanics, QM from propagators, perturbation theory, and the many faces of the propagator. Ch.17 develops field propagators, Feynman propagators, the free scalar propagator, Yukawa force carriers, and the anatomy of a propagator.

Application. The propagator encodes how a particle moves through spacetime. It is the central object of perturbative QFT.

Readings. [L&B Ch.16–17]

1.3.2 Week 7: The S-Matrix and Wick’s Theorem

- L&B Ch.18 covers the S-matrix, interaction representation, scattering, perturbation expansion, and Wick’s theorem. Spend extra time on Wick’s theorem: it converts time-ordered products into normal-ordered products and makes diagram rules possible.

Application. The S-matrix is what one measures in a particle collider. Wick’s theorem is why Feynman diagrams work.

Readings. [L&B Ch.18]

1.3.3 Week 8: Feynman Diagrams and ϕ^4 Theory

- L&B Ch.19 expands the S-matrix, introduces interactions, develops ϕ^4 theory, explains the anatomy of diagrams, symmetry factors, momentum-space calculations, and first scattering examples. Work examples carefully and draw diagrams by hand.

Application. Feynman diagrams are not just cartoons; each is a precise mathematical expression for a scattering amplitude.

Readings. [L&B Ch.19]

1.3.4 Week 9: Scattering Theory and Yukawa Interactions

- L&B Ch.20 covers Yukawa interactions, scattering in $\psi^\dagger\psi\phi$ theory, the transition matrix, invariant amplitudes, and cross-sections. Use this as a consolidation week for propagators, Wick's theorem, and Feynman rules.

Application. The cross-section is the observable. Perturbative QFT ultimately computes cross-sections.

Readings. [L&B Ch.20]

Checkpoint. Midterm checkpoint: draw and evaluate Feynman diagrams for ϕ^4 and Yukawa theories, apply Wick's theorem, and compute a tree-level cross-section.

1.4 Phase 4: Generating Functionals, Statistical Physics, and Path Integrals

The path integral formulation. L&B Parts V–VI; AGD enters as a companion for Green's functions. This phase covers Weeks 10–12.

1.4.1 Week 10: Statistical Physics Interlude and Generating Functionals

- L&B Ch.21 gives a statistical mechanics crash course and introduces sources. Ch.22 develops the generating functional for fields, Green's functions, the Gell-Mann–Low theorem, and diagrammatic calculations. AGD Sections 1–2 provide physical motivation through quasiparticles, liquid helium, superfluidity, and Fermi liquids.

Application. The generating functional $Z[J]$ is central to the path integral approach. Every Green's function is a derivative of Z .

Readings. [L&B Ch.21–22] [AGD §1–2]

1.4.2 Week 11: Path Integrals in Quantum Mechanics and Field Theory

- L&B Ch.23 introduces path integrals in QM, Gaussian integrals, and the SHO propagator. Ch.24 develops functional integrals for fields and the scalar-field generating functional. AGD Section 3 gives a parallel many-body treatment of second quantization.

Application. Feynman's path integral says a particle takes all paths simultaneously, weighted by $e^{iS/\hbar}$.

Readings. [L&B Ch.23–24] [AGD §3]

1.4.3 Week 12: Statistical Field Theory, Broken Symmetry, Coherent States

- L&B Chs.25–28 cover Wick rotation, Euclidean partition functions, finite-temperature Feynman rules, Landau theory, Goldstone modes, the Higgs mechanism, coherent states, Grassmann numbers, and fermionic path integrals. AGD Sections 4–5 give Bose and Fermi gas companions.

Application. Spontaneous symmetry breaking explains both the Higgs mechanism and superconducting order.

Readings. [L&B Ch.25–28] [AGD §4–5]

Checkpoint. Note on AGD: use Abrikosov et al. as a companion rather than the primary text until the Green's-function machinery is comfortable.

Definition 1. A generating functional is a source-dependent object, usually written $Z[J]$, whose functional derivatives generate correlation functions. It is the organizing object behind the path-integral formulation of QFT.

2 Semester B: Renormalization, QED, and Many-Body Field Theory

2.1 Phase 5: Renormalization

Taming ultraviolet divergences. L&B Part VIII; budget extra time for this conceptual section. This phase covers Weeks 13–15.

2.1.1 Week 13: Topology and Quasiparticles (Bridge Week)

- L&B Chs.29–30 cover kinks, vortices, anyons, and Chern–Simons theory as enrichment. Ch.31 introduces renormalization, quasiparticles, the Fermi surface, dressed propagators, and Landau Fermi liquid. AGD Sections 6–7 cover interaction representation and Green’s functions.

Application. Quasiparticles make complicated interacting systems behave like weakly interacting dressed particles.

Readings. [L&B Ch.29–31] [AGD §6–7]

2.1.2 Week 14: Renormalization: The Problem and Its Solution

- L&B Ch.32 explains divergences, counterterms, regularization, and renormalizable theories. Ch.33 shows renormalization in action through propagator corrections, counterterms, renormalization conditions, and vertex functions. AGD Section 8 introduces diagram techniques in many-body language.

Application. Renormalization is not a trick to hide infinities; it describes how physics changes with scale.

Readings. [L&B Ch.32–33] [AGD §8–9]

2.1.3 Week 15: The Renormalization Group

- L&B Ch.34 covers flows in parameter space, RG methods, asymptotic freedom, Anderson localization, and the Kosterlitz–Thouless transition. Ch.35 uses ferromagnetism as an RG tutorial. AGD Section 10 covers Dyson equations and vertex parts.

Application. The RG explains why QED and QCD behave differently at high energies and why critical phenomena are universal.

Readings. [L&B Ch.34–35] [AGD §10]

Checkpoint. Checkpoint: understand why loop integrals diverge, what regularization and renormalization mean physically, and how RG flow describes coupling constants.

2.2 Phase 6: Spinors, QED, and the Dirac Field

Fermions, spinors, and quantum electrodynamics. L&B Part IX. This phase covers Weeks 16–18.

2.2.1 Week 16: The Dirac Equation and Spinors

- L&B Ch.36 covers massless particles, chiral wavefunctions, Dirac and Weyl spinors, and the non-relativistic limit. Ch.37 covers spinor transformations, rotations, boosts, and the four-component Dirac equation.

Application. Dirac predicted antimatter by demanding a relativistic linear wave equation. Spinors are Lorentz-group representations.

Readings. [L&B Ch.36–37]

2.2.2 Week 17: The Quantum Dirac Field and QED

- L&B Ch.38 quantizes the Dirac field, derives Noether currents, fermion propagators, Feynman rules, and local gauge symmetry for fermions. Ch.39 gives a rough guide to QED, including photon propagators, QED Feynman rules, and gauge invariance.

Application. QED is among the most precisely tested theories in science; its rules encode electromagnetic processes.

Readings. [L&B Ch.38–39]

2.2.3 Week 18: QED Scattering and Renormalization of QED

- L&B Ch.40 covers Rutherford scattering, spin sums, the Mott formula, Compton scattering, and crossing symmetry. Ch.41 covers photon-propagator renormalization, dielectric vacuum, running electric charge, vertex corrections, and the electron g -factor. AGD Section 11 introduces finite-temperature Green's functions.

Application. The anomalous magnetic moment of the electron, computed from QED vertex corrections, agrees spectacularly with experiment.

Readings. [L&B Ch.40–41] [AGD §11]

Checkpoint. Checkpoint: write QED Feynman rules, compute a tree-level QED cross-section, and explain the running coupling.

2.3 Phase 7: Many-Body Theory and AGD in Depth

Finite-temperature Green's functions, Fermi liquid theory, and superconductivity. AGD becomes primary. This phase covers Weeks 19–22.

2.3.1 Week 19: Superfluids, Metals, and the Many-Body Problem

- L&B Ch.42 covers Bogoliubov transformations, superfluids, fields, and superfluid currents. Ch.43 treats mean-field theory, Hartree–Fock energy, excitations, electrons and holes, propagators, and RPA. AGD Sections 12–13 introduce finite-temperature perturbation theory and coordinate-space diagrams.

Application. Bogoliubov's transformation diagonalizes the interacting Bose gas and recurs in superconductivity.

Readings. [L&B Ch.42–43] [AGD §12–13]

2.3.2 Week 20: Superconductors and the BCS Theory

- L&B Ch.44 covers Cooper pairs, ground-state energy, bogolons, broken symmetry, and charged superfluid field theory. AGD Sections 14–16 develop finite-temperature momentum-space diagrams and thermodynamic-potential perturbation series.

Application. BCS theory explains superconductivity as a Cooper-pair condensate; the gap is the Higgs mechanism of condensed matter.

Readings. [L&B Ch.44] [AGD §14–16]

2.3.3 Week 21: Fermi Liquid Theory in Depth

- AGD Sections 18–21 cover vertex parts, zero sound, effective mass, Fermi momentum, Bose branches, specific heat, and electron–phonon interactions. L&B Ch.45 on fractional quantum Hall fluids is enrichment.

Application. Landau Fermi liquid theory explains why metals behave like weakly interacting electron gases even with strong interactions.

Readings. [AGD §18–21] [L&B Ch.45 enrichment]

2.3.4 Week 22: Superconductivity Theory in Depth (AGD)

- AGD Sections 32–36 cover microscopic superconductivity, interaction Hamiltonians, the Cooper instability, transition temperature, gap equations at zero and finite temperature, gauge invariance, phonon models, thermodynamics, specific heat, and critical fields.

Application. This is the full microscopic derivation of superconductivity, a landmark application of Green's functions.

Readings. [AGD §32–36]

Checkpoint. Checkpoint: understand Matsubara Green's functions, Dyson's equation, Fermi liquid theory, the BCS gap equation, and the Cooper instability.

2.4 Phase 8: Gauge Theories, Standard Model, and Synthesis

Non-abelian gauge theory, Weinberg–Salam model, advanced topics, and final synthesis. L&B Part XI with AGD parallel reading. This phase covers Weeks 23–26.

2.4.1 Week 23: Non-Abelian Gauge Theory and Yang–Mills

- L&B Ch.46 revisits abelian gauge theory, introduces Yang–Mills theory, non-abelian gauge-field interactions, W_μ dynamics, and symmetry breaking. AGD Sections 28–29 apply Green’s functions to radiation in absorbing media and dielectric constants.

Application. Yang–Mills theory underlies the strong and weak forces and forms the backbone of the Standard Model.

Readings. [L&B Ch.46] [AGD §28–29]

2.4.2 Week 24: The Weinberg–Salam Model and the Higgs Mechanism

- L&B Ch.47 covers electroweak symmetry before breaking, the Higgs field, symmetry breaking, electron mass, photons, and gauge bosons. Ch.48 introduces Majorana fermions, field operators, and Majorana masses. AGD Sections 37–38 cover superconductors in weak electromagnetic fields and near transition temperature.

Application. The Weinberg–Salam model unified weak and electromagnetic forces into electroweak theory.

Readings. [L&B Ch.47–48] [AGD §37–38]

2.4.3 Week 25: Magnetic Monopoles, Instantons, and Topology

- L&B Ch.49 covers Dirac monopoles, Dirac strings, and the ’t Hooft–Polyakov monopole. Ch.50 covers instantons, tunneling, double wells, and false vacuum decay. AGD Sections 40–41 cover non-equilibrium Fermi liquids and analytic properties of vertex parts.

Application. Instantons describe tunneling between topologically distinct vacua and matter for strong CP and baryon-number questions.

Readings. [L&B Ch.49–50] [AGD §40–41]

2.4.4 Week 26: Synthesis, Expository Essay, and Outlook

- Review the full conceptual arc: classical fields to canonical quantization to Feynman diagrams to path integrals to renormalization to gauge theories to many-body Green’s functions. Write the final expository essay and orient toward Peskin–Schroeder, Altland–Simons, and Weinberg.

Application. The final goal is fluency across both particle-physics and condensed-matter field theory languages.

Readings. [L&B review] [AGD §42] [Next: Peskin–Schroeder or Altland–Simons]

Checkpoint. Final assessment: write a 4–6 page expository essay deriving and explaining a result such as the BCS gap equation, QED Compton scattering, the running coupling in QED, or Bogoliubov’s transformation.

Theorem 1 (Noether’s Theorem). *Every continuous symmetry of the action determines a conserved current. In QFT this is the bridge between spacetime/internal symmetries and conservation laws such as energy, momentum, charge, and angular momentum.*

3 Final Assessment

- Write a 4–6 page expository essay deriving and explaining one result in detail.
- Suggested topics include the BCS gap equation, QED Compton scattering, the running coupling in QED, Bogoliubov’s transformation for superfluids, or the Higgs mechanism.

4 Topics Covered

- **Foundations:** Lagrangians, classical fields, occupation numbers, second quantization, canonical quantization, symmetries, and Noether's theorem.
- **Perturbative QFT:** propagators, Green's functions, S-matrix, Wick's theorem, Feynman diagrams, ϕ^4 theory, Yukawa theory, and scattering cross-sections.
- **Path integrals and renormalization:** generating functionals, functional integrals, Wick rotation, Grassmann variables, spontaneous symmetry breaking, counterterms, regularization, and renormalization group flow.
- **Spin and gauge theory:** Dirac equation, spinors, Dirac field, QED, gauge invariance, non-abelian Yang–Mills theory, electroweak symmetry breaking, Majorana fermions, monopoles, and instantons.
- **Many-body field theory:** quasiparticles, finite-temperature Green's functions, Dyson equations, Fermi liquid theory, superfluids, superconductors, BCS theory, and condensed matter applications of QFT.

5 Recommended & Further Readings

- Peskin and Schroeder, *An Introduction to Quantum Field Theory*: the standard graduate QFT reference for particle physics.
- Altland and Simons, *Condensed Matter Field Theory*: the natural continuation of AGD into modern condensed matter.
- Weinberg, *The Quantum Theory of Fields*, Vols. I–III: the definitive full treatment, best read after L&B.
- Coleman, *Aspects of Symmetry*: legendary lectures on instantons, solitons, and the large- N expansion.